

Question Paper

Exam Date & Time: 23-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Advanced Computer Networks [CSE_5151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A rescue team utilises UAVs to establish communication links in disaster areas. Analyse how UAV networks differ from MANETs and VANETs in handling mobility and topology changes. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) A logistics company operates delivery drones across multiple urban zones. Apply a suitable UAV network topology to ensure efficient communication and justify your choice. (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) A cloud service provider experiences delays due to manual configuration of numerous network devices. Explain with appropriate reasoning, how Software-Defined Networking (SDN) automation addresses the challenges of traditional network management. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) An organisation wants to integrate its existing legacy network into an SDN environment. Apply Cisco ACI's policy extension mechanisms to achieve unified management across old and new systems. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A financial firm wants to consolidate servers to reduce power consumption and idle capacity. Analyse how compute virtualisation transforms server utilization and operational efficiency. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) A global IT enterprise plans to virtualise both compute and storage resources. Apply the principles of Software Defined Storage to unify heterogeneous storage systems. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) A mobile user experiences sudden drops in video quality while watching over 4G. Apply DASH which adapts video bitrates to maintain seamless streaming under fluctuating bandwidth. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) A video-conferencing system struggles with lip-sync mismatches. Apply RTP timestamping and synchronisation methods to align audio and video streams. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) A router must distribute bandwidth fairly between video and file-transfer traffic. Evaluate the effectiveness of Weighted Fair Queuing (WFQ) in maintaining fairness and delay control. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) A healthcare data system requires continuous uptime despite regional fiber faults. Explain how self-healing ring architectures restore connectivity automatically to prevent service interruption (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_5151:Advanced Computer Networks

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	15
5	5	CO 5	5
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_5151:Advanced Computer Networks

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	6	30(60%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)